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Eliminating the Negative Pore Synergy in Hierarchical Porous Metal-Organic Frameworks for Isomer Separation

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Hierarchical porous metal-organic frameworks (MOFs) integrating micro- and mesopores hold promise for advanced separations but often suffer from understudied negative pore synergy, where interconnected pores compromise selectivity and diffusion. Using the zirconium-based porous coordination network (PCN) PCN-608 as a model, we identify that rapid analyte translocation between meso-hexagonal and micro-triangular channels induces chaotic diffusion, undermining separation efficiency. A channel-isolation strategy is then developed via solvent-assisted installation of barrier ligands (BDC/NH_2BDC) at interconnecting windows. PCN-608-BDC with isolated pores exhibit 8-13 times higher diffusion coefficients for xylene isomers than the pristine PCN-608 monitored by inverse gas chromatography (IGC). Molecular dynamics simulations confirm the restriction of cross-pore migration and the acceleration of diffusion kinetics in PCN-608-BDC. The PCN-608-BDC shows obviously better performance than the pristine material as GC stationary phases and breakthrough adsorbents in separation xylene isomers. All PCN-608 series with a micro-mesoporous mixed structure exhibit excellent xylene uptake. Similar improvements in separation performance are also observed for NU-1000 with isolated pores, validating the universality of the phenomena. By balancing pore connectivity and active site availability, this work establishes channel isolation as a generalizable design principle for optimizing hierarchical MOFs to eliminate the negative pore synergy, offering simultaneous enhancements in selectivity, and stability for gas separations.

Introduction

Microporous materials (pore diameter <2 nm) and mesoporous materials (2–50 nm) have both emerged as pivotal platforms in separation science due to their distinct yet complementary structural advantages.[1-5] Microporous frameworks, characterized by densely distributed active sites and strong host-guest interactions, excel in selective molecular recognition and precise separation of small analytes (Fig. 1a).[6, 7] In contrast, mesoporous architectures enable rapid molecular transport through enlarged diffusion channels, significantly boosting the diffusion rates and adsorption capacities (Fig. 1b).[8, 9] The integration of these hierarchical pore systems has been extensively explored to synergize their merits, aiming to achieve simultaneous enhancements in molecular selectivity, diffusion kinetics, and adsorption performance.[10-19] Such hybrid configurations are particularly attractive for applications demanding both high-resolution separation and efficient mass transfer, exemplified by gas separation.[20-23]

Despite these theoretical benefits, when micropores and mesopores are interconnected through unrestricted lateral pore windows, analyte molecules may diffuse multi-directionally rather than following the most energetically or kinetically favorable transport pathway. Similar diffusion interference has been experimentally and computationally observed in other porous systems. For example, Liu *et al.* showed that intersecting channels in H-ZSM-5 caused diffusion anisotropy, reducing the effective transport rate.[24] Similarly, cross-channel exchange in Zn_2(dobpdc) had been reported to slow the diffusion relative to strictly one-dimensional pathways.[25] The interpore diffusion in Cu-based MOF has been discovered to govern the mass uptake of the whole MOF particle.[26] These studies demonstrate that pore-connectivity pattern also plays a critical role in determining transport behavior. In the hierarchical MOFs, when analyte molecules traverse interconnected microporous and mesoporous networks with minimal interaction time, neither pore system can fully exploit its inherent advantages (Fig. 1c). The micropores fail to establish selective adsorption equilibrium, while the mesopores lose their kinetic superiority in molecular transport. This negative pore synergy phenomenon remains critically understudied. We posit that the main challenges lie in the synthetic complexity of precisely

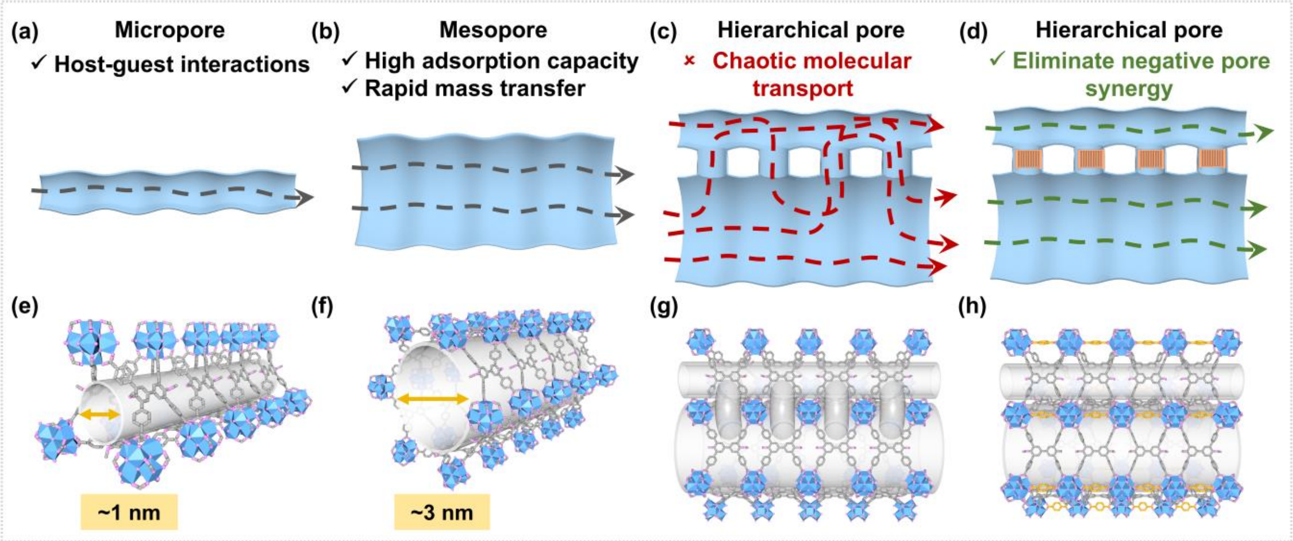

Fig. 1| The illustration of different channels and diffusion paths in PCN-608. Schematic diagrams of **a** micropore, **b** mesopore, **c** interconnected hierarchical pore, and **d** isolated micropores and mesoporous networks, respectively. The illustration of the **e** micro-triangular channel, **f** meso-hexagonal channel, and **g** interconnected pore networks of PCN-608, respectively. **h** The illustration of isolated two pore systems employing barrier ligands. Zr_6 cluster, blue polyhedra; O, pink; C of parent framework linker (TPCB), gray; C of barrier ligand (BDC), golden; hydrogen atoms omitted for clarity.

engineered hierarchical structures, the technical limitations in quantifying transient molecular-pore interactions, and the prevailing research focus on amplifying positive synergy. Bridging this knowledge gap is essential for the rational design of next-generation separation media, as unmitigated negative pore cooperativity could limit their practical applications.

The zirconium-based porous coordination network (PCN) PCN-608, featuring a ***csq*** topology[27, 28], is selected as an ideal model for studying and controlling negative pore synergy. Its structure consists of two interconnected pore networks: meso-hexagonal channels (~3 nm) and micro-triangular channels (~1 nm). The side-wall pore windows (8.2 Å × 13.2 Å), lined with unsaturated Zr sites, serve as high-affinity adsorption hotspots that guide guest molecules through the apertures.[26, 29] Such dual-channel transport drives rapid analyte shuttling between the two pore types, which can give rise to negative synergistic effects (Fig. 1e-g).[30-32] Importantly, the zirconium clusters in PCN-608 feature four open metal sites, which enable the post-synthetic attachment of barrier ligands.[33-37] These ligands can selectively block interchannel openings, effectively narrowing the accessible sub-window size to approximately 6 Å, thereby preventing rapid analyte transfer between the two channels (Fig. 1d, h). Furthermore, tuning the functional groups of these barrier ligands[27, 38] enables systematic investigation into how interfacial interactions influence the negative pore synergy. Thus, PCN-608 offers a precisely controllable platform to decouple negative pore synergy between two pore types and explore the corresponding mechanisms.

Herein, the intertranslocation between different channels was first investigated through Grand Canonical Monte Carlo (GCMC) simulations with para-xylene (pX) as the probe molecule. The simulations revealed that pX molecules rapidly shuttled between the interconnected hexagonal channels and triangular channels of PCN-608, resulting in disordered diffusion pathways. To suppress this behavior, the interconnections between the two pore systems were selectively blocked via a solvent-assisted ligand incorporation strategy, employing benzene-1,4-dicarboxylic acid (BDC) and 2-aminobenzene-1,4-dicarboxylic acid (NH_2BDC) as the barrier ligands (Fig. 2a). The thermodynamic and kinetic properties of pristine PCN-608, PCN-608-BDC, and PCN-608-NH_2BDC were then systematically evaluated by inverse gas chromatography (IGC) using xylene isomers as probes. Thermodynamic analysis of xylenes revealed that PCN-608-BDC exhibited weaker host-guest interactions than pristine PCN-608. Blocking the open metal sites significantly attenuated the thermodynamic interactions, supporting that the anticipated confinement-driven enhancement in adsorption enthalpy (ΔH) upon eliminating intermolecular translocation did not occur. Expectedly, PCN-608-NH_2BDC displayed stronger interactions with xylenes than PCN-608-BDC, attributed to the additional hydrogen-bonding capability of the NH_2 groups within the confined triangular channels. Notably, kinetic studies further demonstrated that the diffusion coefficients (D_s) of pX in PCN-608-BDC and PCN-608-NH_2BDC were markedly higher than in pristine PCN-608. The diffusion rate constants (K_{sap}) of pX derived from single-component vapor adsorption experiments and the self-diffusion coefficient values of pX calculated via molecular dynamics (MD) simulations also corroborated these improvements. These findings indicate that isolating the translocation between hexagonal and triangular channels effectively eliminated negative pore synergy. In practical implications, all three PCN-608 series exhibited high xylene adsorption due to their hierarchical porous structures. PCN-608-BDC exhibited much higher GC separation resolution for all tested isomer mixtures than pristine PCN-608. Moreover, PCN-608-BDC achieved significantly better separation capability than PCN-608 in binary-phase breakthrough experiments for xylene-isomers separation. To generalize this strategy, the same BDC modification was applied to NU-1000, a widely studied MOF with ***csq*** topology. NU-1000-BDC likewise outperformed its pristine framework in separation performance, confirming that eliminating negative pore synergy universally improves the performance of hierarchical pore systems. Collectively, these results establish that eliminating negative pore synergy is an effective strategy for optimizing molecular selectivity and transport kinetics in advanced separation materials.

Results and discussion

Characterizations of PCN-608, PCN-608-BDC and PCN-608-NH_2BDC. The crystallinity and phase purity of the synthesized MOFs were confirmed using powder X-ray diffraction (PXRD). Experimental PXRD patterns of PCN-608 showed an excellent agreement with the simulated diffractograms, confirming its structural integrity. Moreover, both PCN-608-BDC and PCN-608-NH_2BDC exhibited PXRD patterns nearly identical to that of PCN-608, indicating that the incorporation of the BDC and NH_2BDC linkers did not alter the underlying framework structure (Fig. 2b). Scanning electron microscopy (SEM) was used to investigate the morphology of the three materials. The SEM images revealed that all three materials consistently maintained a hexagonal prism morphology, with particle lengths predominantly centered at approximately 220 nm. This observation suggested that the linker incorporation process had minimal impact on the particle size distribution (Fig. 2c and Supplementary Fig. 1). These results collectively demonstrated that the introduction of BDC or NH_2BDC linkers into PCN-608 did not compromise the crystallinity, phase purity, or morphological features of the parent framework.

To further confirm the successful installation of the

BDC and NH_2BDC linkers, proton nuclear magnetic resonance (^1H NMR) spectra of three digested MOF samples were collected. The aromatic protons of the BDC linker exhibited a chemical shift at approximately 8.09 ppm, which overlapped with signals from the 4,4'-dimethoxybiphenyl 3,3',5,5'-tetra(phenyl-4-carboxylic acid) (TPCB) ligand. Quantitative analysis revealed that the final BDC-to-TPCB ratio was 0.51:1 in PCN-608-BDC (Supplementary Fig. 2 and 3). PCN-608 features two distinct types of open metal sites. The first type comprises pairs of opposing open metal sites located at the 8.2 Å side-wall pore windows, which can be bridged by a single BDC ligand, thereby isolating adjacent channels. In contrast, the second open metal site type, located within the mesopores, allows only monodentate dangling coordination of BDC ligands, and this connection is intrinsically unstable and can be readily removed[27] (Supplementary Fig. 4). Theoretical calculations predict a stable maximum TPCB-to-BDC ratio of 1:0.5 for fully coordinated PCN-608-BDC. The experimentally observed ratio (1:0.51) closely aligned with this theoretical value, confirming high-occupancy installation of BDC linkers at the targeted unsaturated metal sites. Unlike BDC, NH_2BDC exhibited two chemical shifts at 7.10 ppm and 7.43 ppm (Supplementary Fig. 5). The NH_2BDC-to-TPCB ratio in PCN-608-NH_2BDC was calculated as 0.57:1, which was also close to the theoretical ratio, also supporting the effective and precise installation of the NH_2BDC ligands.

To assess the permanent porosity of the three MOFs, nitrogen adsorption-desorption experiments were conducted at 77 K under atmospheric pressure. As shown in Fig. 2d, all three materials displayed similar characteristic Type IV isotherms, indicative of their structural similarity. The Brunauer-Emmett-Teller (BET) surface areas of PCN-608-BDC and PCN-608-NH_2BDC,

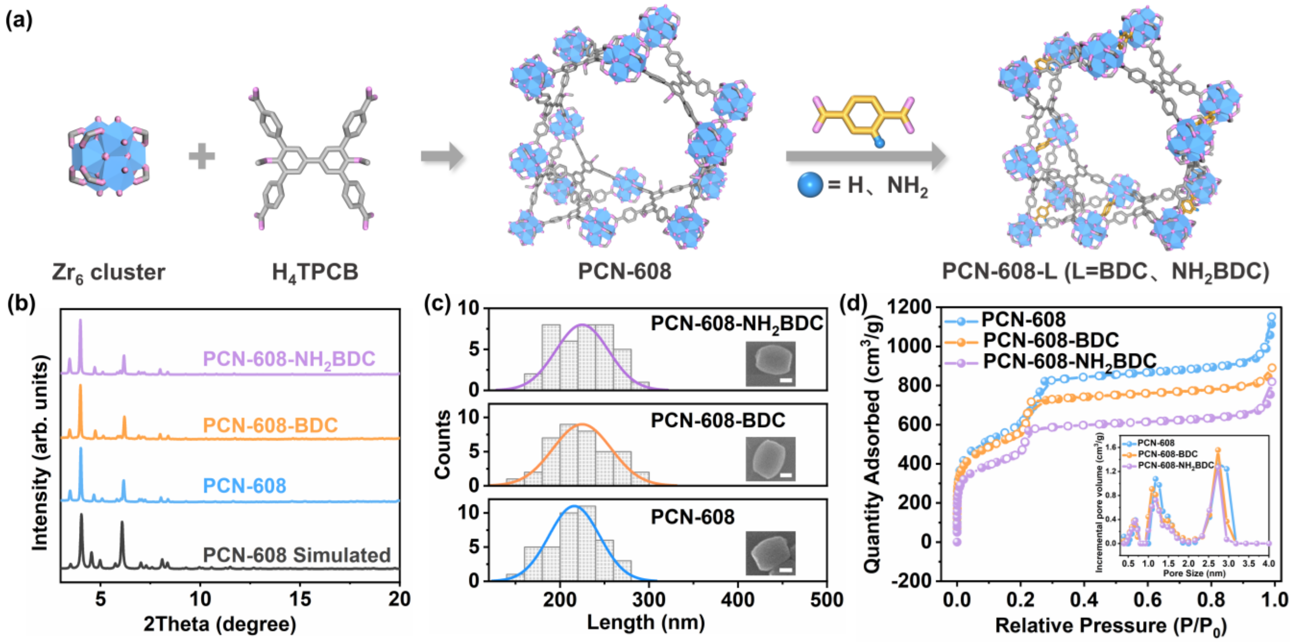

Fig. 2| Schematic illustration and characterization of PCN-608, PCN-608-BDC and PCN-608-NH_2BDC. a The construction of PCN-608 from Zr_6 cluster and H_4TPCB ligand. Schematic illustration of linker installation on the prismatic pore windows of PCN-608. Zr_6 cluster, blue polyhedra; O, pink; C of parent framework linker (TPCB), gray; C of barrier ligand (BDC, NH_2BDC), golden; N, light blue; hydrogen atoms omitted for clarity. **b** PXRD patterns of PCN-608 series. **c** Statistical histograms of particle length for PCN-608 series from SEM images. The scale bar in each SEM image is 100 nm. **d** N_2 adsorption-desorption isotherms at 77 K and pore size distribution (DFT method) of PCN-608 series.

calculated from the nitrogen adsorption data, were determined to be 1982 and 1606 m^2/g, respectively, both lower than that of PCN-608 (2115 m^2/g). This reduction was attributed to the partial occupation of unsaturated metal sites by the installed barrier ligands.[39] The pore size distribution profiles derived from density functional theory (DFT) analysis of the nitrogen adsorption data revealed that all three materials retained the dual pore characteristics of PCN-608. However, the positions of the incremental pore volume for both micropore and mesopore regions in PCN-608-BDC and PCN-608-NH_2BDC exhibited slight shifts toward smaller pore sizes compared with PCN-608. This subtle contraction was likely due to the spatial encroachment imposed by the installed BDC and NH_2BDC linkers. These results suggested that while the overall pore framework remained intact, the local pore environments were modestly modulated by the linker incorporation. Similar shifts in both micropore and mesopore pore-size distribution profiles were also observed in argon adsorption–desorption measurements conducted at 87 K, further supporting the reliability of these conclusions (Supplementary Fig. 6).

Discover the Intermolecular Translocation of Analytes in PCN-608. The direct experimental verification of negative pore synergy remains challenging due to the inherent difficulty in isolating and independently evaluating the contributions from hexagonal and triangular channels within a single MOF. However, intermolecular shuttling between these interconnected pores systems could indeed induce varying degrees of negative pore synergy. To unravel this phenomenon, GCMC and MD simulations were synergistically employed to probe pX transport dynamics within PCN-608[40, 41]. GCMC simulations, performed on the rigid MOF framework at 303 K using Lennard-Jones (L-J) potentials and the Universal Force Field (UFF), revealed that pX molecules preferentially adsorbed not only within the channels but also accumulated at the interconnecting pore windows (Supplementary Fig. 7), suggesting potential transport behaviors. A similar phenomenon was observed in a previous study of NU-1000 with the same ***csq*** topology, where molecules were found to preferentially localize near the Zr_6 clusters before diffusing to other regions.[42] Subsequent MD simulations under the constant-volume-and-temperature (NVT) ensemble further revealed the localization of multiple pX molecules at the interconnecting pore windows. Analysis of pX distribution snapshots in PCN-608 from 0 ps to 450 ps and then to 1000 ps demonstrated distinct molecular orientations at these windows, indicating intermolecular translocation. This observation indicated that diffusion in PCN-608 did not proceed along a single dominant pathway but instead followed multiple lateral routes, resulting in erratic and non-directional molecular trajectories. Such multichannel hopping extended the effective diffusion length and reduced transport efficiency, contrary to the expected benefits of hierarchical porosity. Furthermore, the results showed that many pX molecules accumulated around the unsaturated Zr nodes at the side-wall pore windows, indicating that the strong adsorption affinity of these exposed Zr sites further diverted diffusing molecules toward lateral apertures. This repeated lateral deflection interrupted directional transport and reinforced the overall chaotic diffusion behavior. The combined computational insights linked pore

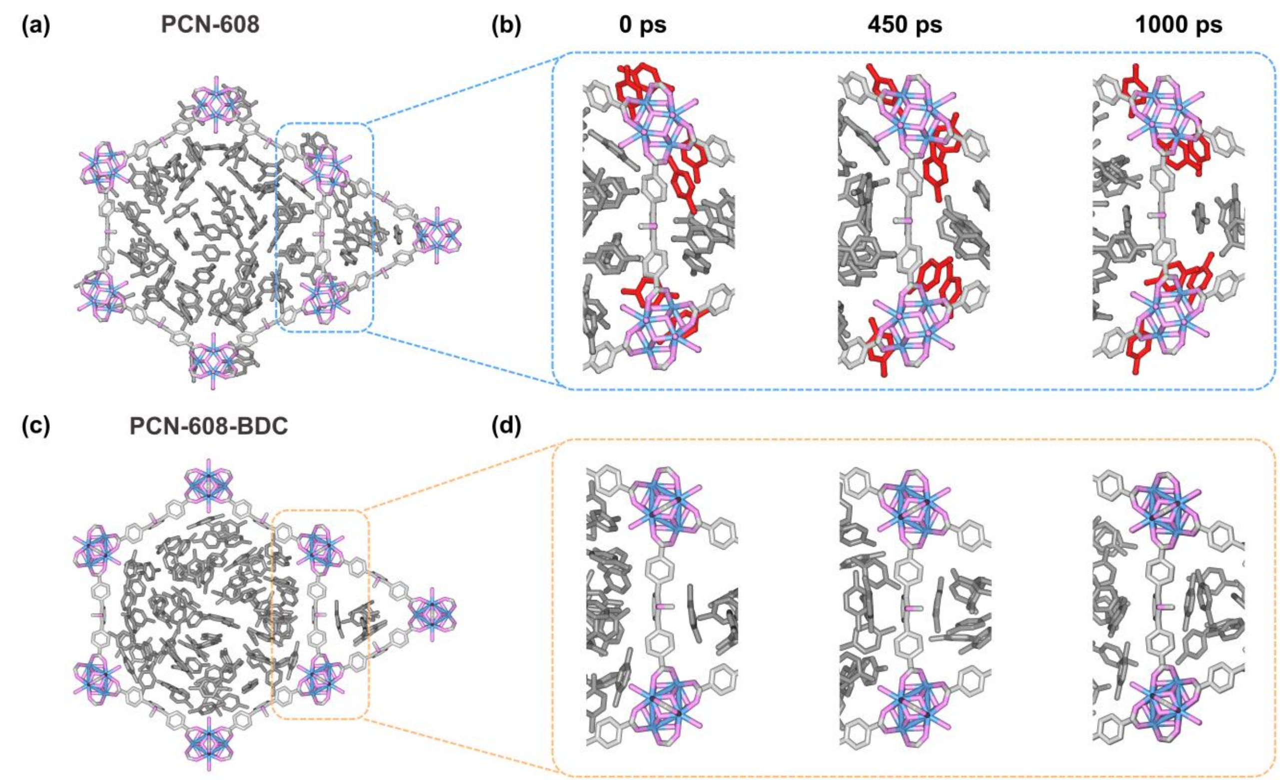

Fig. 3| Simulation of the intertranslocation between different channels. Molecule distributions of p-xylene (pX) molecules in **a** PCN-608, and **c** PCN-608-BDC simulated by Grand Canonical Monte Carlo (GCMC) simulations at 303 K. Enlarged view of pX distributions near the interconnecting pore windows of **b** PCN-608, and **d** PCN-608-BDC simulated by molecular dynamics (MD) simulations at different time. Zr, blue; O, pink; C of parent framework linker (TPCB), light gray; C of pX, dark gray; hydrogen atoms omitted for clarity. The pX molecules at the interconnecting pore windows were highlighted in red.

interconnectivity to chaotic diffusion, providing mechanistic evidence for negative pore synergy arising from unregulated intermolecular translocation.

Eliminate the Negative Pore Synergy. To prevent intermolecular translocation and eliminate negative pore synergy in PCN-608, BDC and NH_2BDC linkers were introduced via a solvent-assisted ligand incorporation strategy to selectively block the interconnecting pore windows, yielding PCN-608-BDC and PCN-608-NH_2BDC, respectively. Similar to our approach in PCN-608, both GCMC and MD simulations were performed on PCN-608-BDC with pX as the target molecule to examine whether channel isolation altered molecular transport behavior. In PCN-608-BDC, pX molecules were exclusively confined within the channels, with no accumulation observed at the pore windows, indicating successful blockage of interchannel connections (Fig. 3c and Supplementary Fig. 8). Consistently, MD simulations revealed that pX molecules maintained well-defined channel trajectories throughout the entire simulation timescale, with no evidence of lateral migration (Fig. 3d). The complete absence of molecular intertranslocation at the side-wall pore windows indicated that the micropore and mesopore networks became structurally decoupled after BDC installation, thereby restoring directional transport behavior.

Inverse gas chromatography (IGC), a technique uniquely sensitive to host-guest interactions and molecular diffusion,[43, 44] was then applied to probe how these structural modifications influenced analyte confinement and transport dynamics within the MOF channels. To comprehensively characterize the system, we systematically evaluated both thermodynamic adsorption parameters (e.g., ΔH and ΔG) and kinetic diffusion coefficients (D_s) using xylene isomers as model analytes.

The ΔH values for xylene isomers were calculated via the van't Hoff equation (Fig. 4a, Supplementary Fig. 9 and Table 1). Compared with pristine PCN-608, the modified PCN-608-BDC and PCN-608-NH_2BDC columns exhibited less negative ΔH values for all three isomers, indicating weaker thermodynamic interactions between the MOFs and the analytes. These counterintuitive results contrasted with the commonly observed confinement-enhanced interactions and can be attributed to the occupation of active sites on Zr_6 clusters by the BDC and NH_2BDC linkers, which directly diminishing the available adsorption sites for xylenes (Fig. 4b)[45, 46]. Consequently, the anticipated thermodynamic enhancement from pore confinement was overshadowed by the loss of active sites. This phenomenon underscored the trade-off between pore isolation and active site availability in governing host-guest thermodynamics. Notably, PCN-608-NH_2BDC exhibited slightly more negative ΔH values than PCN-608-BDC due to supplementary N–H···π interactions introduced by the additional NH_2 groups. Moreover, the ΔH values for the three xylene isomers remained nearly identical on each column, highlighting the limited thermodynamic differentiation among them across all three materials. Despite this, the sequences of ΔH values for pX, meta-xylene (mX), and ortho-xylene (oX) varied among different materials, indicating distinct interaction types with the xylene isomers. Notably, the Gibbs free energy differences (ΔG) values among the three xylene

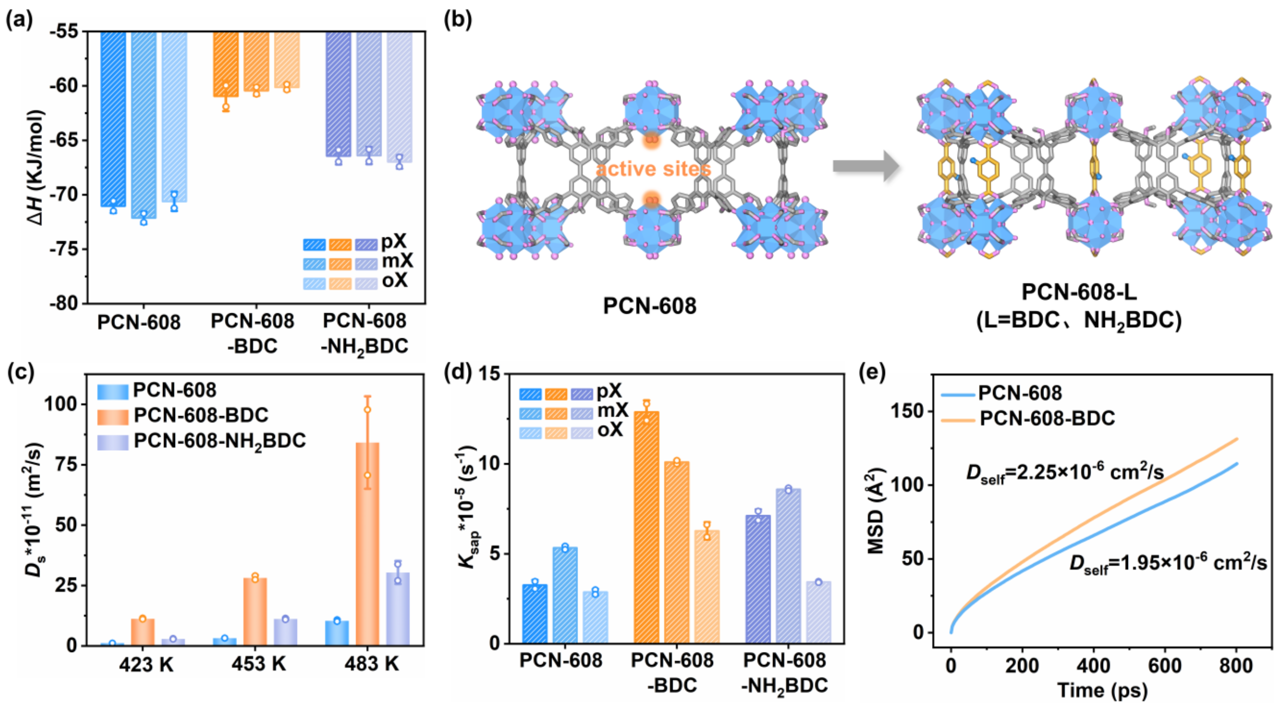

Fig. 4| The thermodynamic and kinetic properties of PCN-608 series. a Adsorption enthalpy (Δ*H*) of xylene isomers in three PCN-608-series MOF-coated columns based on van't Hoff plots. **b** Schematic representation of the unsaturated Zr sites. Zr_6 cluster, blue polyhedra; O, pink; C of parent framework linker (TPCB), gray; C of barrier ligand (BDC, NH_2BDC), golden; N, light blue; hydrogen atoms omitted for clarity. **c** Diffusion coefficient (D_s) values of pX in PCN-608-series MOF coated columns obtained from Golay equation curves at 423 K, 453 K, and 483 K, respectively. **d** The calculated diffusion rates of xylene in PCN-608 series at 303 K. **e** MD-derived self-diffusion rates of pX in PCN-608 and PCN-608-BDC at 303 K. For all the bar charts in this figure, bars represent the mean of replicate experiments (n = 2), and error bars indicate the standard deviation (SD).

isomers were nearly indistinguishable in pristine PCN-608 but became substantially differentiated in PCN-608-BDC and PCN-608-NH_2BDC (Supplementary Table 2). This result indicated that thermodynamic discrimination between isomers was significantly enhanced. We considered that the strong adsorption affinity of the unsaturated Zr nodes resulted in the nonspecific adsorption of the isomers, diminishing thermodynamic discrimination of the framework. In contrast, PCN-608-BDC and PCN-608-NH_2BDC effectively passivate these nonspecific adsorption sites, thereby enhancing the discrimination.

The D_s values of xylenes in PCN-608, PCN-608-BDC, and PCN-608-NH_2BDC were calculated at 483 K using the Golay equation (Supplementary Fig. 10-13 and Table 3). The results demonstrated that both modified frameworks exhibited significantly enhanced diffusion coefficients compared with pristine PCN-608. For PCN-608-BDC, the D_s values for pX, mX, and oX reached 84.2, 95.0, and 59.9×10^{-11} m^2/s, respectively, which are 8 to 13 times higher than those in PCN-608 (10.5, 7.4, and 4.6×10^{-11} m^2/s, respectively). The steric hindrance introduced by NH_2 groups in PCN-608-NH_2BDC slightly attenuated this enhancement relative to PCN-608-BDC, although both modified materials outperformed the parent framework. Crucially, this kinetic improvement persisted across a broad temperature range, with the D_s values for pX in the modified materials remaining elevated even at lower temperatures (Fig. 4c and Supplementary Fig. 14-16). The superior diffusion kinetics observed in PCN-608-BDC and PCN-608-NH_2BDC arose from blocked interchannel pore windows, which enforced uniform diffusion pathways by restricting analyte migration between micro- and mesopores. When mesopores dominated molecular transport, xylenes encountered fewer diffusion barriers and diffused more rapidly. This structural isolation eliminated chaotic cross-pore shuttling, thereby mitigating negative pore synergy and optimizing kinetic performance.

To distinguish whether the enhanced diffusion arises primarily from channel isolation rather than from reduced thermodynamic adsorption strength, two additional materials, PCN-608-BA (BA=benzoic acid) and PCN-608-BDC-BA, were designed. Quantitative analysis showed that the TPCB-to-BA ratio in PCN-608-BA was 2:2.8 (Supplementary Fig. 17). Owing to its small molecular size, BA could not fully isolate the side-wall windows of PCN-608. PCN-608-BA exhibited ΔH values that were even less negative than those of PCN-608-BDC, yet its D_s value showed no appreciable increase relative to PCN-608 when using pX as the target analyte (Supplementary Fig. 18, 19, and Table 1, 3). This finding indicated that the enhancement in diffusion could not be attributed primarily to weakened thermodynamic adsorption. To further investigate the individual contributions of residual unsaturated Zr sites within the mesopores of PCN-608-BDC, BA was introduced as a monodentate modulator onto the remaining open metal sites, aiming to further suppress thermodynamic interactions while preserving structural isolation. Despite efforts to achieve full saturation of the two accessible sites per mesopore, ^{1}H NMR analysis revealed a maximum loading of approximately 1.37 BA ligands per Zr_6 cluster. Thermodynamic data revealed a less negative ΔH value for pX compared with PCN-608-BDC, indicating weaker host–guest interactions due to additional passivation of strong adsorption sites. Correspondingly, the D_s value of pX in this material increased relative to pristine PCN-608. However, owing to the steric hindrance of BA, the D_s value of pX in PCN-608-BDC-BA remained lower than that in PCN-608-BDC (Supplementary Fig. 20-22 and Table 1, 3). These results collectively demonstrated that the observed enhancement in molecular diffusion was predominantly governed by the elimination of chaotic cross-channel transport through structural pore isolation, rather than solely by weakening host–guest interactions. This highlighted the critical role of directional confinement in modulating analyte mobility within hierarchical MOFs.

To further characterize diffusion behavior under high molecular loading, single-component vapor adsorption experiments were conducted for xylene isomers in PCN-608, PCN-608-BDC, and PCN-608-NH_2BDC until adsorption saturation was reached. The diffusion rate constants for xylene isomers at 303 K were obtained by fitting the kinetic adsorption data using the linear driving force (LDF) model (Supplementary Fig. 23-25).[47, 48] For PCN-608-BDC, the K_{sap} values reached 1.29×10^{-4}, 1.01×10^{-4}, and 6.27×10^{-5} s^{-1} for pX, mX, and oX, respectively, which were 1.9- to 4.0-fold higher than those in pristine PCN-608 (3.25×10^{-5}, 5.33×10^{-5}, and 2.87×10^{-5} s^{-1}, respectively). Consistent with the observations in IGC, the K_{sap} values in PCN-608-NH_2BDC increased less significantly than those in PCN-608-BDC but remained higher than those in PCN-608 (Fig. 4d). This kinetic superiority was further demonstrated by the shorter saturation time of xylene isomers in PCN-608-BDC and PCN-608-NH_2BDC than in PCN-608. For example, pX reached adsorption saturation at 20 min in PCN-608-BDC, 23 min in PCN-608-NH_2BDC and 36 min in PCN-608, respectively (Supplementary Fig. 26). These results supported the superior diffusion kinetics of analytes in PCN-608-BDC relative to in PCN-608 even under elevated molecular loading.

Complementing experimental findings, GCMC and MD simulations were employed to elucidate the diffusion behavior[40] of pX in PCN-608 and PCN-608-BDC at 303 K. The MD results showed that pX could readily transfer from the hexagonal channels to the triangular channels in PCN-608 (Fig. 3 and Supplementary Fig. 7). In contrast, after the modification of BDC as barrier ligands at the interconnecting pore windows, no pX molecules were observed to shuttle back and forth between the two types of channels in PCN-608-BDC (Supplementary Fig. 8). The self-diffusion coefficient of pX, derived from the slope of mean square displacement (MSD) curves, increased from 1.95×10^{-6} cm^2/s in pristine PCN-608 to 2.25×10^{-6} cm^2/s in PCN-608-BDC (Fig. 4e), confirming enhanced diffusion kinetics in the modified framework. These computational results aligned with the experimental observations, demonstrating that blocking interconnecting pore windows effectively accelerates analyte transport and thereby eliminates negative pore synergy.

Practical Applications of Eliminating the Negative Pore Synergies. The modified PCN-608-BDC and PCN-608-NH_2BDC frameworks displayed distinct thermodynamic adsorption and kinetic diffusion behaviors compared with pristine PCN-608, motivating the practical application of eliminating the negative pore

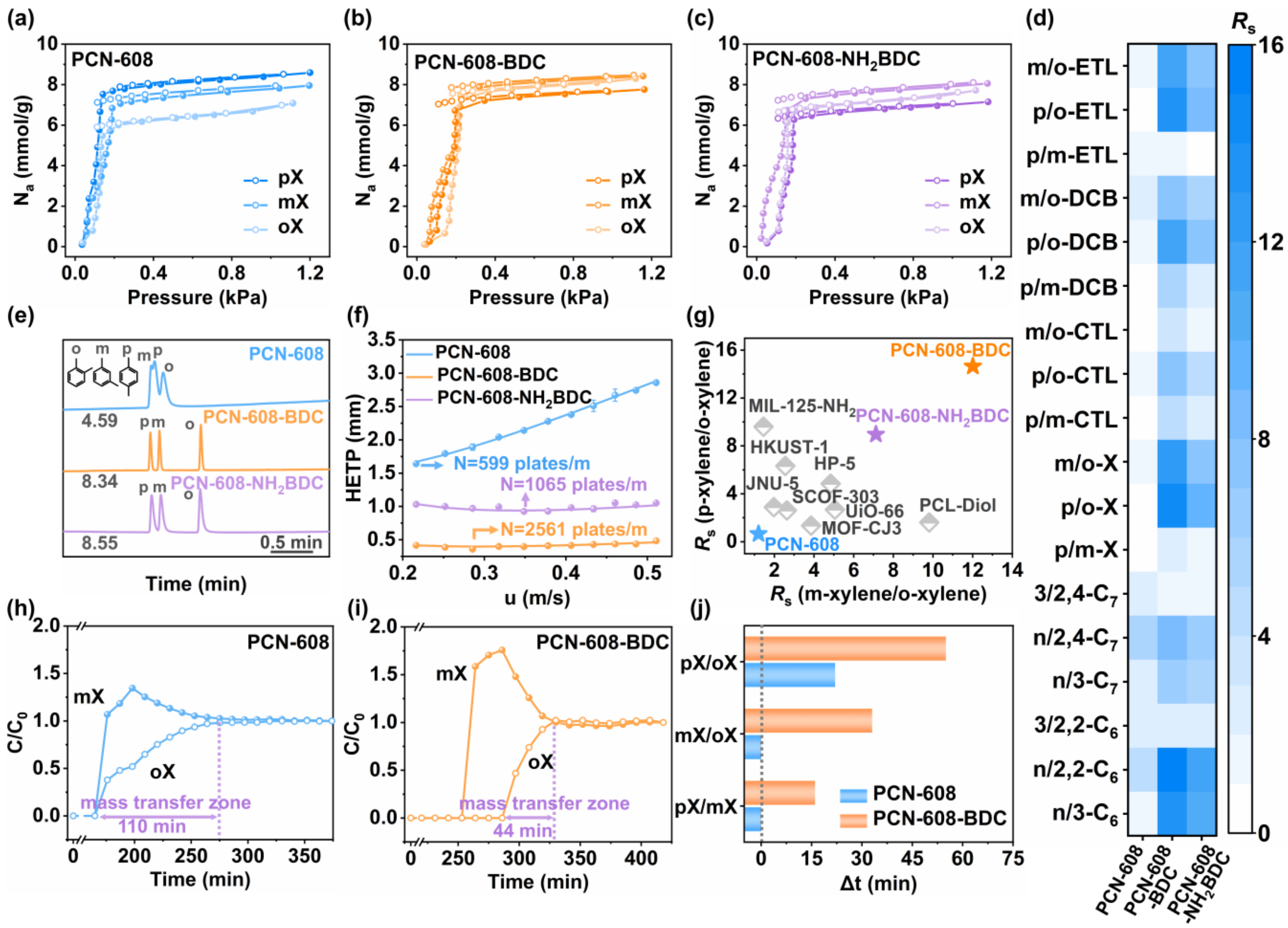

Fig. 5| Practical applications of eliminating the negative pore synergies. Single-component vapor adsorption isotherms of xylene isomers on **a** PCN-608, **b** PCN-608-BDC, and **c** PCN-608-NH_2BDC at 303 K, respectively. **d** Separation resolution of disubstituted benzene isomers and alkane isomers on PCN-608 series coated columns. ETL, DCB, and CTL represent ethyltoluene, dichlorobenzene, and chlorotoluene, respectively. n-C_7, 3-C_7, 2,4-C_7 represent n-heptane, 3-methylhexane, and 2,4-dimethylpentane, respectively. n-C_6, 3-C_6, 2,2-C_6 represent n-hexane, 3-methylpentane, and 2,2-dimethylbutane, respectively. **e** Gas chromatograms of xylene isomers separated by PCN-608, PCN-608-BDC, and PCN-608-NH_2BDC coated columns. The number on the left under each chromatogram represents the elution time of the first peak. **f** Golay curves of PCN-608 series coated columns with pX as the target molecule at 453 K. **g** Resolution (R_s) values of pX/oX and mX/oX on the PCN-608 series and other materials coated columns.[43, 49-55] Experimental breakthrough curves for equimolar vapor-phase mixtures of mX/oX passing through the columns packed with **h** PCN-608, and **i** PCN-608-BDC. **j** The difference in breakthrough time for an equimolar binary mixture of xylene isomers at 303 K passing through the PCN-608 and PCN-608-BDC columns.

synergy. Single-component vapor adsorption isotherms of xylene isomers were collected at 303 K for PCN-608, PCN-608-BDC and PCN-608-NH_2BDC powder samples (Fig. 5a-c). The adsorption capacities of pX, mX, and oX on PCN-608 reached up to 8.59, 6.96, and 7.08 mmol g^{-1}, respectively, at 1.1 kPa. Both modified materials still exhibited high adsorption capacities for xylene. Remarkably, all three PCN-608 variants outperformed previously reported porous materials in xylene adsorption,[47, 56-62] underscoring the superior performance of these hierarchically structured MOFs (Supplementary Fig. 27 and Table 4). This balance between high adsorption capacity and tunable pore connectivity highlighted their potential for advanced separation applications,[63] in which structural modifications eliminate negative pore synergy without sacrificing intrinsic adsorption efficiency.

Subsequently, the three PCN-608 series materials were dynamically coated onto capillary columns and evaluated as GC stationary phases. Chromatograms were acquired under optimized separation conditions specific to each chromatographic column and isomer mixture. The PCN-608-coated column failed to resolve xylene isomers, whereas the PCN-608-BDC-coated column achieved baseline separation with resolution (R_s) values of 14.64 (pX/oX) and 12.01 (mX/oX), outperforming most reported stationary phases and the commercial HP-5 column (Fig. 5e, g).[43, 49-51] Although the PCN-608-NH_2BDC-coated column exhibited slight peak tailing due to steric hindrance and additional interactions from NH_2 groups, it still effectively separated xylene isomers. Notably, the elution order of the isomers shifted after linker modification, consistent with differences in steric hindrance. Pore isolation effectively suppressed chaotic inter-pore diffusion, thereby enforcing directional analyte transport within one-dimensional microporous channels. This structural confinement enhanced kinetic discrimination, enabling more precise resolution of isomers based on subtle differences in molecular geometry and diffusivity. Both PCN-608-BDC and PCN-608-NH_2BDC columns demonstrated narrower peaks and higher R_s values for alkane and substituted benzene isomers compared with PCN-608 (Fig. 5d, e, Supplementary Fig. 28-32, and Table 5-6). Notably, the PCN-608-NH_2BDC column exhibited lower performance than the PCN-608-BDC column, highlighting that eliminating negative pore synergy in hierarchical MOFs through different post-modification strategies can lead to distinct trade-offs between thermodynamic interactions and kinetic diffusion of analytes. The additional material PCN-608-BA exhibited poor separation performance, while PCN-608-BDC-BA demonstrated markedly improved separation behavior, comparable to that of PCN-608-NH_2BDC (Supplementary Fig. 33-35, and Table 7). Therefore, optimizing the barrier ligands was crucial for achieving optimal separation performance across different applications.

Subsequently, the column efficiency was further assessed using pX at 453 K. The PCN-608-BDC and PCN-608-NH_2BDC columns exhibited significantly lower heights equivalent to a theoretical plate (HEPT) than PCN-608 across a wide linear velocity range. Specifically, PCN-608-BDC achieved an optimal efficiency of 2561 plates/m at 0.27 m/s, while PCN-608-NH_2BDC reached 1065 plates/m at 0.35 m/s, far exceeding that of PCN-608 (599 plates/m at 0.22 m/s) (Fig. 5f). These results further underscored the benefits of eliminating negative pore synergy in achieving high GC performance. Long-term stability tests further revealed severe peak broadening in PCN-608 columns after two weeks, whereas the PCN-608-BDC column retained stable performance after daily frequent usage under high-temperature conditions for 16 months. Remarkably, columns re-prepared using six-month-aged PCN-608-BDC powder maintained effective xylene separation (Supplementary Fig. 36-39), demonstrating that pore window isolation not only enhances kinetic and thermodynamic performance but also significantly improves material durability for practical applications.

To further assess the practical implications of eliminating negative pore synergy, PCN-608 and PCN-608-BDC were tested as adsorbents for binary xylene isomer separations via breakthrough experiments, given the industrial significance of these compounds (Fig. 5h-j, and Supplementary Fig. 40 and 41). For an equimolar mX/oX mixture at 303 K and a flow rate of 45 sccm, both isomers co-eluted simultaneously through the PCN-608-packed column. In contrast, PCN-608-BDC achieved sequential elution, with mX breaking through first and oX emerging after a 33-minute delay. The mass transfer zone from breakthrough to saturation further highlighted the accelerated kinetics, in that oX required 110 minutes to saturate PCN-608 but only 44 minutes in PCN-608-BDC. Similar improvements were observed for other binary mixtures, where PCN-608-BDC exhibited larger differences in analyte breakthrough time and narrower mass-transfer zones compared with PCN-608. Both materials followed the same elution order (pX < mX < oX), and all breakthrough curves displayed a pronounced "roll-up" effect, with transient C/C_0 values exceeding 1.0, indicative of competitive adsorption dynamics.[64-67] These results further demonstrated that isolating pore networks via linker modification not only enhanced separation selectivity but also optimized mass transfer efficiency, positioning PCN-608-BDC as a promising candidate for practical industrial xylene purification.

Universality of Eliminating the Negative Pore Synergy. To further validate the universality of eliminating negative pore synergy in enhancing hierarchical MOF performance, this strategy was extended to NU-1000,[68, 69] a zirconium-based MOF sharing the same ***csq*** topology as PCN-608 but differing in organic ligands. Using a postsynthetic modification analogous to that employed for PCN-608, BDC linkers were incorporated into NU-1000 to yield NU-1000-BDC (Fig. 6a). Structural integrity was confirmed by PXRD, which showed retained crystallinity and phase purity for both pristine NU-1000 and modified NU-1000-BDC. SEM imaging revealed a consistent rice-grain morphology for both materials. (Supplementary Fig. 42 and 43) Both materials were coated onto capillary columns and evaluated as GC stationary phases (Fig. 6b, c, and Supplementary Fig. 44, 45, and Table 8). The pristine NU-1000-coated column displayed poor resolution for C_9 isomers, with severe peak overlap, while NU-1000-BDC achieved baseline separation. Similarly, for dichlorobenzene isomers, NU-1000-BDC exhibited well-resolved peaks, whereas NU-1000 suffered from pronounced peak tailing. We recognized that the improved separation performance also resulted from the enhanced mass transfer caused by the pore isolation and the reduced nonspecific adsorption arising from the occupation of unsaturated Zr nodes. These improvements mirrored the performance trends observed in PCN-608-BDC, underscoring the generality of pore isolation in optimizing hierarchical MOFs. The consistent success across two structurally distinct MOF platforms highlighted the broad applicability of this strategy for designing precision separation systems, where controlled pore connectivity eliminated negative pore synergy while simultaneously amplifying kinetic and thermodynamic advantages.

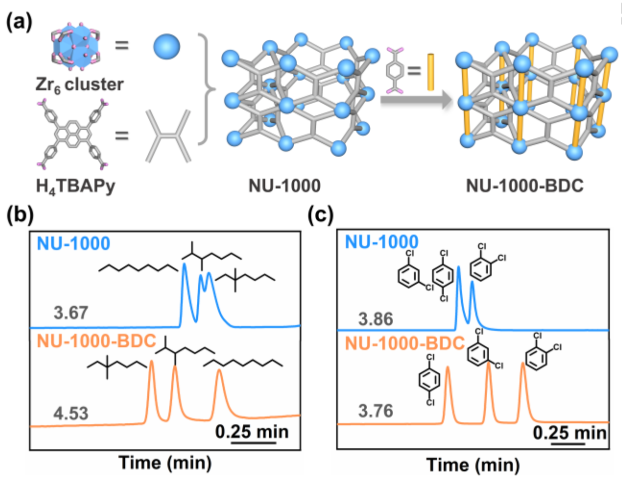

Fig. 6| Universality of eliminating the negative pore synergy. a The construction of NU-1000 from Zr_6 cluster and H_4TBAPy ligand. The post-synthetic linker installation towards NU-1000-BDC. Gas chromatograms of NU-1000 and NU-1000-BDC coated columns for the separation of **b** C_9 isomers, and **c** dichlorobenzene isomers. The number on the left above each chromatogram represents the elution time of the first peak.

In summary, we demonstrated that eliminating negative pore synergy in hierarchical MOFs through targeted pore isolation significantly enhances separation performance. By incorporating BDC and NH_2BDC barrier ligands into PCN-608, the interconnected micro-triangular and meso-hexagonal channels were effectively decoupled, suppressing chaotic intermolecular translocation of analyte. Both modified frameworks exhibited markedly improved diffusion kinetics for xylene isomers compared with pristine PCN-608, as confirmed by IGC. BDC ligands occupied metal adsorption sites in PCN-608, thereby prioritizing kinetic enhancement by minimizing cross-pore migration and achieving faster mass transfer efficiency with weaker thermodynamic interactions with analytes. Similarly, PCN-608-NH_2BDC possessed weaker interactions with analytes and faster mass transfer than PCN-608. The introduction of NH_2 groups imparted additional N–H···π interactions and steric hindrance into the framework, resulting in slower analyte diffusion and stronger host-guest affinity relative to PCN-608-BDC. Although both modified materials showed slightly reduced xylene adsorption capacities compared with pristine PCN-608, they outperformed the parent framework in separation performance as GC stationary phases and in breakthrough experiments. Extending this strategy to NU-1000 further validated the universality of eliminating negative pore synergy in enhancing the separation performance of frameworks, with modified materials consistently surpassing their parent structures in GC applications. Critically, the modified materials exhibited exceptional long-term stability (>16 months), far exceeding that of pristine PCN-608 (~2 weeks), underscoring their industrial viability. By rationalizing pore connectivity and active site engineering, this work provides a blueprint for designing hierarchical MOFs that balance selectivity, transport kinetics, and durability, advancing their utility in high-resolution separations.

Methods

Synthesis of PCN-608. PCN-608 was synthesized according to the previous work with a few changes.[69] Specifically, 10 mg $ZrCl_4$ and 120 μL acetic acid (AA) were dissolved in 2 mL DMF in a 4 mL glass vial. The solution was heated at 373 K for 1 h. After cooling to room temperature, 20 mg H_4TPCB ligand, 160 μL AA, and 200 μL deionized water were added to the mixture. The resulting mixture was sonicated until it became clear and transparent, followed by heating at 393 K for 24 h. The white powder was collected by centrifugation and washed thoroughly with DMF and ethanol three times, respectively.

Synthesis of PCN-608-BDC. Approximately 15 mg as-synthesized PCN-608 was soaked in 2 mL DMF with the addition of 25 mg H_2BDC dissolved in 4.6 mL DMF. The mixture was stirred at 348 K for 24 h. The powder of PCN-608-BDC was collected by centrifugation and soaked in fresh DMF for 3 days to remove uncoordinated linkers and monodentate dangling BDC species, and then soaked in ethanol for 3 days to remove DMF in the MOF channel.

Synthesis of PCN-608-NH_2BDC. The powdery PCN-608-NH_2BDC was synthesized by a similar procedure as PCN-608-BDC using 30 mg NH_2BDC instead of H_2BDC. The material processing steps are the same as those in PCN-608-BDC.

GCMC Simulation. The GCMC simulation was performed by

Materials Studio according to the experimental temperature and pressure of MOFs coated columns in the IGC experiments. In the GCMC simulations, Lennard-Jones (L-J) potential and Universal Force Field (UFF) were used to simulate the non-bonding interactions of the MOF system. The MOFs were considered rigid. For the p-xylene molecule, the parameters were adopted from the Trappe force field. Electrostatic potential (ESP) charges were applied to MOFs and p-xylene based on periodic density functional theory (PDFT) calculations. For each thermodynamic state, the number of equilibration steps and production steps were set to 5×10^6 each.

MD Simulation. The MD simulations were performed by Materials Studio according to experimental conditions. The initial configurations of the system, the charges, and the L-J parameters of atoms were taken from the results of GCMC simulations. The electrostatic interactions were handled by the Ewald summation method, while Van der Waals interactions were described by the atom-based summation approach. The cutoff for Ewald summation and Van der Waals interactions was set a 12.5 Å and accuracy was set to 0.001 kcal/mol. The constant-volume-and-temperature (NVT) ensemble was used under 303 K controlled by Nose-Hoover thermostat. The time step and total simulation time were set as 1.0 fs and 1000 ps, respectively. The self-diffusion coefficient of p-xylene in MOFs was calculated from the slope of the mean square displacement (MSD) of p-xylene.

Data availability

The data that support the conclusions of this study are either presented in the paper or its Supplementary Information. Source data are provided with this paper.

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Author contributions

Z.-Y.G. and M.X. conceived the idea and supervised the research. J.-J.L. and M.X. performed the synthesis, characterization, and GC experiments and discussed the results. S.-S.M., Y.-K.K. and H.Y. assisted in the GC experiments. W.L. and C.-Y.R. assisted in the vapor adsorption measurements and breakthrough experiments. Z.-Y.G., J.-J.L., and M.X. discussed the experimental data and wrote the paper. All authors have approved the final version of the manuscript.

[#]These authors contributed equally.

Competing interests

The authors declare no competing interests.

Additional information

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Editorial Summary

Here authors show that isolating interconnected channels in metal-organic frameworks eliminates inefficient molecular shuttling, greatly improving diffusion and separation performance. The strategy enhances isomer separation and material stability.

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